

UNIT 5: Fuels, Combustion & Corrosion

1. Fuels & Calorific Value

A fuel is a combustible substance containing Carbon and Hydrogen that releases large amounts of heat energy upon burning in air.

Gross (HCV) vs Net (LCV) Calorific Value:

- HCV / Gross CV: Total heat produced when 1 unit fuel is burned completely and products are cooled to 15°C (steam condenses).

- LCV / Net CV: Heat produced when steam is allowed to escape without condensing.

$LCV = HCV - (0.09 \times H \times 587) \text{ cal/g}$ (where H = % Hydrogen in fuel).

Bomb Calorimeter:

Used to measure HCV of solid and liquid fuels.

$$HCV = [(W + w)(t_2 - t_1) - \text{Corrections}] / m \text{ cal/g}$$

where W = mass of water, w = water equivalent of calorimeter, m = mass of fuel.

Dulong's Theoretical Formula:

$$HCV = \frac{1}{2} 100 [8080 C + 34500 (H - O/8) + 2240 S] \text{ kcal/kg}$$

2. Lubricants & Mechanisms

Substances applied between moving metal surfaces to reduce friction, heat, and wear.

Lubrication Mechanisms:

1. Hydrodynamic (Fluid-Film): Thick liquid layer ($> 1000 \text{ \AA}$) completely separates moving parts under low load/high speed.
2. Boundary Lubrication: Thin monolayer absorbed on surface under high load/low speed when fluid film breaks.
3. Extreme Pressure (EP): Under extreme load/high temp, EP additives (organic compounds of S, P, Cl) react with metal surface forming high-melting protective metallic films.

3. Key Lubricant Properties

- Viscosity Index (VI): Measure of change of viscosity with temperature. High VI = minimal change (ideal for engines!).
- Flash Point: Minimum temp at which fuel gives off vapor that ignites momentarily with a spark.

4. Corrosion & Cathodic Protection

Corrosion is the gradual destruction of metal due to chemical/electrochemical reaction with environment.

Types of Electrochemical Corrosion:

- Galvanic Corrosion: When two dissimilar metals are in contact in electrolyte, more active metal acts as Anode and corrodes.
- Pitting Corrosion: Localized attack forming deep pin-holes due to breakdown of protective oxide surface film.

Cathodic Protection Methods:

1. Sacrificial Anode Method: Structure to be protected (e.g., steel pipeline) is connected by wire to a more reactive metal anode (Zn, Mg bar). The sacrificial anode corrodes, protecting the steel!
2. Impressed Current Cathodic Protection (ICCP): External DC current is applied in opposite direction to neutralize corrosion current.

5. Quick Exam Revision Summary

- LCV: $LCV = HCV - 0.09 H \times 587$.
- Dulong Formula: $HCV = \frac{1}{2}100 [8080C + 34500(H - \frac{O}{8}) + 2240S]$.
- Viscosity Index: Resistance to viscosity change with temp.
- Sacrificial Anode: Zn/Mg attached to protect iron.